Lantern: Honest Amplification for Endangered-Language Revival

A Decidable Attestation Constraint that Turns a Probabilistic Language Model into a Provably Non-Confabulating Documentation Engine*

S. Kodithyala† Areeb Hirani Vinay Pulavarthy

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Abstract

Roughly half of the world's approximately seven thousand languages are projected to fall silent within a century. The dominant framing treats this as a sociopolitical inevitability; we argue that it is, in substantial part, a *tooling* problem, because the cost of documenting a language and building pedagogy for it has historically been measured in linguist-years. Large language models could collapse that cost, but applied naively to an extremely-low-resource language they *confabulate*: they produce fluent, plausible, and wrong material that pollutes a fragile record. This paper presents Lantern, a deployed system that inverts the usual role of the model: rather than originating linguistic knowledge, the model only *restructures and amplifies* knowledge a community already holds, under a hard honesty constraint enforced in code. We give a formal account of that constraint as a decidable *attestation* predicate over a tokenized corpus, and prove a zero-hallucination soundness theorem: every artifact that reaches a learner is composed solely of tokens attested by cited community evidence. We show a laundering-impossibility corollary (an invented form can never enter the admissible language) and a monotonicity proposition (the admissible language is non-decreasing in the corpus while the hallucination count is identically zero, independent of corpus size). The filter is linear-time, deterministic, and independent of the probabilistic model. On two cited corpora (56 phrases) the engine induces 48 vocabulary items, 7 grammar patterns, 22 flashcards, and 7 practice sentences with 0 hallucinated tokens reaching a learner and 48/48 vocabulary items carrying a citation, recomputed independently at a live metrics endpoint. We argue the constraint that appears limiting is precisely the unlock.

Keywords: endangered languages; computational language documentation; LLM hallucination; faithfulness; attestation; constrained generation; neuro-symbolic guardrails; grammatical induction; spaced repetition.

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1 Introduction

A language does not die for want of a dictionary; it dies when the last person stops speaking it. But *revival*, the practical project of bringing a new generation to fluency, depends on materials:

*Source code: github.com/areebhirani-beep/lantern. Live system: lantern-cyan.vercel.app. Every quantitative claim in this paper is reproducible against the repository and the live `/api/metrics` endpoint; Table 2 maps each formal construct in the text to its exact source location.

†Corresponding author. Correspondence concerning this article may be addressed to the author.

glossaries, reference grammars, graded lessons, and audio. Producing these is slow, specialist work. A field linguist may spend years documenting a single language, and there are far more endangered languages than there are linguists or years (Austin and Sallabank, 2011; Moseley, 2010). The result is triage: a handful of languages receive sustained attention, most receive none, and time runs out. Of the roughly seven thousand languages in use, a large fraction are transmitted to few or no children (Eberhard et al., 2021; Joshi et al., 2020). This is the leverage point that motivates the present work. If the marginal cost of turning *what a community remembers* into *teachable material* can be reduced from years to minutes, the arithmetic of triage inverts, and revival becomes tractable for the long tail of languages that documentation has never reached.

1.1 The extremely-low-resource confabulation failure

The naive application of a large language model (LLM) to this problem fails, and fails dangerously. A language with fifty speakers has essentially no presence in any pretraining corpus. Asked to "teach" such a language, a model will *confabulate*, emitting fluent, plausible, and incorrect forms (Ji et al., 2023; Maynez et al., 2020). For a critically endangered language this is worse than useless: it pollutes the documentary record and disrespects the people who still carry the knowledge. A trained linguist catches the error immediately; a learner, by construction, cannot. Any system that aspires to help here must therefore make a guarantee that ordinary generation cannot: that *nothing fabricated ever reaches the learner*. The central technical question of this paper is how to obtain such a guarantee from a probabilistic model that offers no such promise on its own.

1.2 The inversion: amplify, do not originate

The originating thesis of LANTERN is an inversion of the usual "AI as oracle" framing. We do not ask the model to *know* a language, which, for an extremely-low-resource language, is impossible, but to *reason over a community's own data*, which is tractable and is precisely what field linguists do. Concretely, the model receives a small *seed corpus* of phrases with glosses, performs interlinear analysis *on that corpus only*, must *cite* the phrases that attest each lexical or grammatical claim, and may recombine attested material according to patterns it actually found, but may *never* introduce a form from outside the corpus. The honesty constraint is not a limitation to be engineered around; it is the design that makes the system work, and the same mechanism that keeps the engine honest is the mechanism that makes it general, because it depends on the community's data rather than the model's memory. We call this regime *honest amplification*: a community-in-the-loop amplifier that makes a small amount of human knowledge teach at scale without ever fabricating.

1.3 Contributions

This paper makes the following contributions.

C1 (formal honesty constraint).
We formalize honesty for generated language as a decidable *admission* predicate over a tokenized, citation-verified corpus (Section 3), separating *originated* from *restructured* knowledge.

C2 (soundness theorem).
We prove a zero-hallucination soundness theorem (Theorem 1), a laundering-impossibility corollary (Corollary 1), and a monotonicity proposition (Proposition 1).

C3 (constrained induction pipeline).
We give a constrained induction-and-assembly pipeline (Algorithm 1) coupling forced structured tool-use to a hard post-hoc attestation filter (Algorithm 2).

C4 (self-recomputing evaluation).
We report an evaluation whose central safety property is *independently recomputed* from the same tokenizer at a live endpoint, rather than self-reported (Section 5).

C5 (deployed system and flywheel).
We describe a deployed, provider-agnostic web system whose corpus, and hence whose admissible language, grows monotonically as learners contribute.

Roadmap. Section 2 positions LANTERN against related work. Section 3 fixes notation and defines hallucination, attestation, and the honesty constraint. Section 4 presents the engine: induction, the attestation filter, the soundness results, a complexity and determinism analysis, and the fail-soft architecture and flywheel. Section 5 reports the evaluation. Sections 6 through 8 discuss generalization, limitations and ethics, and conclusions.

2 Related Work

Endangered-language and low-resource NLP. Language documentation has long been framed around portability and reuse of primary data (Bird and Simons, 2003), and recent work foregrounds the political and ethical asymmetries of building language technology *for* rather than *with* communities (Bird, 2020). Quantitative surveys of linguistic diversity in NLP show that the overwhelming majority of languages are effectively absent from the resources and benchmarks that drive the field (Joshi et al., 2020; Eberhard et al., 2021), and reference works catalogue both the scale of endangerment and its vitality gradations (Moseley, 2010; Austin and Sallabank, 2011). LANTERN is squarely a tool for this regime; its distinguishing move is to treat the community's small cited corpus, not a pretrained model, as the sole source of lexical content.

Unsupervised morphological and grammatical induction. A long line of work induces linguistic structure from raw or lightly annotated text: unsupervised morphology from distributional regularities and minimum-description-length objectives (Goldsmith, 2001), and grammar induction for dependency and constituency structure (Klein and Manning, 2004). LANTERN uses an LLM as a flexible inducer over a tiny corpus, but unlike classical inducers, which only ever recombine observed material, a generative model can also *invent*. The contribution is not the inducer but the hard constraint that confines its output to attested material.

Hallucination and faithfulness. Hallucination is a well-documented failure mode of neural generation (Ji et al., 2023), and faithfulness has been studied closely in summarization, where fluent outputs routinely assert facts unsupported by the source (Maynez et al., 2020). Most mitigations are *soft*, training objectives, reranking, or decoding heuristics that reduce but do not eliminate unsupported content. LANTERN instead enforces a *hard*, decidable property as a post-hoc filter, trading expressive freedom for a guarantee.

Retrieval-augmentation and attribution. Retrieval-augmented generation conditions a model on retrieved evidence to improve factuality (Lewis et al., 2020). Such methods make hallucination *less likely* by supplying context, but the generator may still produce tokens absent from the retrieved evidence; attribution is typically a soft, learned signal. LANTERN differs in kind: every emitted token must lie in the closure of the cited corpus under verification, checked symbolically after generation, so the guarantee does not depend on the model attending to the evidence.

Neuro-symbolic guardrails and constrained decoding. Constrained decoding restricts generation to a grammar or lexicon, and neuro-symbolic guardrails wrap a model in symbolic checks. LANTERN's filter is a guardrail of this family, but it is applied at the level of *attestation by cited evidence* rather than mere membership in a fixed lexicon, and it verifies citations (catching both invented forms and miscitations), which is what yields the soundness theorem of Section 4.3.

Spaced repetition and computer-assisted language learning. The delivered course uses the SM-2 spaced-repetition scheduler, whose spacing optimization is well studied (Woźniak and Gorzelańczyk, 1994), embedded in a standard computer-assisted language learning loop. This component is not novel; it is included to show that the honest output is delivered as a usable course rather than a static report.

3 Problem Formulation

We now fix notation and define the properties the engine must guarantee. The definitions are deliberately model-agnostic: they concern only strings, tokens, and citations, so that the resulting guarantee holds regardless of which (or whether any) probabilistic model produced a candidate.

3.1 Corpus, normalization, and tokenization

Let Σ be the orthographic alphabet and Σ^* the set of finite strings over it. Let $V \subseteq \Sigma^*$ be the token vocabulary. A *corpus* is a finite multiset

$$C = \{p_1, \ldots, p_n\}, \qquad p = (\mathrm{tgt}_p, \mathrm{gloss}_p), \quad \mathrm{tgt}_p \in \Sigma^*,$$

where tgt_p is the target-language surface string of phrase p and gloss_p its meaning. Normalization is a map $\nu : \Sigma^* \to \Sigma^*$ that case-folds and strips punctuation; in the implementation it lowercases and removes punctuation, and macron-bearing letters and syllabary characters fold correctly under this map. Writing $\mathrm{split}(s)$ for the whitespace segmentation of s, the tokenizer is

$$\tau(s) \;=\; \{\, \nu(w) : w \in \mathrm{split}(s) \,\} \;\setminus\; \big(\mathrm{GAP} \cup \{\varepsilon\}\big), \tag{1}$$

where ε is the empty string and GAP is a fixed set of *gap tokens*, the ellipsis and dash markers $\{\,\ldots,\ \ldots,\ \text{—},\ \text{-}\,\}$, used to write discontinuous morphemes (for example the Māori continuous frame *e … ana*). Gap tokens are notational scaffolding, not words, and are removed so that they neither enter the attested set nor are required of a generated form. Throughout, τ is a single canonical implementation shared by the engine and by the independent evaluator (Section 5); there is no second, drifting copy.

3.2 Attested set, verification, and admission

The *corpus attested set* is the set of all tokens that occur in any phrase target,

$$A_C \;=\; \bigcup_{p \in C} \tau(\mathrm{tgt}_p). \tag{2}$$

This is the ground truth: a token belongs to A_C exactly when the community's cited corpus actually exhibits it.

A *candidate vocabulary item* is a pair $v = (f_v, E_v)$ with surface form $f_v \in \Sigma^*$ and cited evidence $E_v \subseteq C$, the set of phrases the inducer claims attest v. The *verification predicate* accepts v exactly when its form is nonempty and every token of the form occurs in one of its *own* cited evidence phrases:

$$\mathrm{Ver}(v) \iff \tau(f_v) \neq \varnothing \ \wedge\ \tau(f_v) \subseteq \bigcup_{p \in E_v} \tau(\mathrm{tgt}_p). \tag{3}$$

Because the right-hand union ranges over the *cited* phrases rather than over all of C, (3) rejects both wholly invented forms and forms that are real but *miscited* (cited to a phrase that does not contain them). An empty form fails by the first conjunct.

The *attestation basis* extends A_C by the tokens of every *verified* form,

$$A \;=\; A_C \ \cup \bigcup_{v\colon \mathrm{Ver}(v)} \tau(f_v). \tag{4}$$

Finally, for any generated surface form s (a practice sentence or a flashcard answer), the *admission predicate* is

$$\mathrm{Adm}(s) \iff \tau(s) \neq \varnothing \ \wedge\ \tau(s) \subseteq A. \tag{5}$$

The engine emits an artifact s *if and only if* $\mathrm{Adm}(s)$ holds.

3.3 Hallucination and the honesty constraint

Definition 1 (Hallucinated token)**.** A token w is *hallucinated* with respect to a corpus C if it is presented to a learner yet does not lie in the closure of C under cited verification, that is, w is neither in A_C nor a token of any verified form whose own citation places it in A_C.

Definition 2 (Honesty constraint / soundness)**.** A documentation engine is *sound* (equivalently, satisfies the *honesty constraint*) if every artifact it emits satisfies Adm, and consequently presents no hallucinated token to a learner.

The remainder of the paper constructs an engine that provably satisfies Definition 2, and reports that on its shipped corpora the realized count of hallucinated tokens reaching a learner is 0.

4 The Honest Amplification Engine

Lantern is a deployed Next.js / TypeScript web application (lantern-cyan.vercel.app); its engine is provider-agnostic across several LLM backends and fixtures, on the principle that *the model proposes and the code disposes.* The engine runs a single loop with four stages and one invariant: *seed*, *induce*, *generate (assemble)*, and *learn*, with a contribution step that feeds new phrases back to the seed. Figure 1 shows the loop, with the cite-or-reject filter as its load-bearing gate.

4.1 Induction under forced structured tool-use

The induction stage casts the model as a field linguist constrained to a *single* small corpus. Under a field-linguist system prompt and a forced structured tool call (`report_induction`), the model performs interlinear alignment of words to glosses, hypothesizes morphology from minimal pairs,

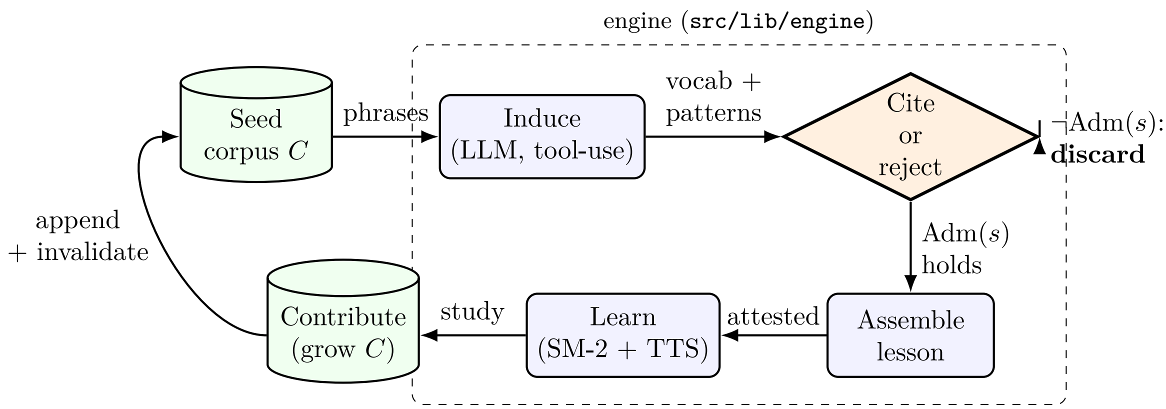

Figure 1: The Lantern loop. A cited seed corpus C is induced into vocabulary and grammar patterns by an LLM under forced structured tool-use; the *cite-or-reject* gate admits only artifacts s with $\mathrm{Adm}(s)$ true (Eq. 5) and **discards** any artifact containing an unattested token; admitted material is assembled into a lesson and delivered through spaced repetition; learner contributions append to C and invalidate the cached induction, so the next visit re-derives richer material. The corpus, and hence the admissible language, grows monotonically (Proposition 1).

and proposes controlled recombinations of attested material. It returns (i) a *vocabulary bank*, each item $v = (f_v, E_v)$ carrying the phrase identifiers E_v that attest it, and (ii) *grammar patterns*, each carrying corpus examples and a confidence. The tool output is validated against a runtime `zod` schema before any use; malformed output is rejected outright. Algorithm 1 states the loop. Two properties matter for soundness. First, the model is required to *cite*; an item without evidence cannot be verified. Second, induction is deterministic for unmodified seed corpora via a seed-pinned fixture, and falls back to a verified fixture on any error, so the pipeline never throws (fail-soft; Section 4.5).

4.2 The attestation filter

The assembly stage applies the attestation filter of Eqs. (2)–(5). The basis A is built once per turn from the corpus tokens together with *only* the tokens of verified vocabulary forms; both candidate *flashcard answers* and candidate *practice sentences* are then filtered by Adm, and any candidate carrying even a single unattested token is dropped before it can reach a learner. The filter is dependency-free: it calls no model and no database. Algorithm 2 gives the procedure.

4.3 Soundness

We now prove that the engine satisfies the honesty constraint of Definition 2.

Theorem 1 (Zero-hallucination soundness)**.** *For every admitted artifact s and every token $t \in \tau(s)$, t is attested by C: either $t \in A_C$, or $t \in \tau(f_v)$ for some v with $\mathrm{Ver}(v)$, in which case t itself occurs in a phrase cited by v and hence $t \in A_C$. Therefore no hallucinated token (Definition 1) ever reaches a learner.*

Proof. Suppose s is admitted, so $\mathrm{Adm}(s)$ holds. By (5), $\tau(s) \subseteq A$, and by (4), $A = A_C \cup \bigcup_{v:\,\mathrm{Ver}(v)} \tau(f_v)$. Fix $t \in \tau(s)$; then $t \in A$. If $t \in A_C$ we are done. Otherwise $t \in \tau(f_v)$ for some v with $\mathrm{Ver}(v)$. By the verification predicate (3), $\tau(f_v) \subseteq \bigcup_{p \in E_v} \tau(\mathrm{tgt}_p)$, and since $E_v \subseteq C$ we have

Algorithm 1: INDUCE + ASSEMBLE (one loop turn)

Input: seed corpus $C = \{p_1, \ldots, p_n\}$; language id ℓ
Output: lesson $L = (cards, practice)$ of admitted artifacts

$\mathcal{R} \leftarrow$ RUNINDUCTION(C, ℓ)
  where RUNINDUCTION returns the seed-pinned fixture if C is an unmodified seed corpus,
  else the validated live LLM tool output, and on *any* error the verified fixture (never throws)
assert $\mathcal{R}$ conforms to the `zod` induction schema; otherwise **reject**
$(Vocab, Patterns) \leftarrow \mathcal{R}$
$A \leftarrow$ BUILDBASIS$(C, Vocab)$ `// Eq. (3)-(4); see Alg. 2`
$cards \leftarrow [\,]$; $practice \leftarrow [\,]$
foreach *candidate flashcard answer a proposed from Vocab* **do**
  if ADMIT(a, A) **then**
    append a to *cards*
foreach *candidate practice sentence s recombined from Vocab, Patterns* **do**
  if ADMIT(s, A) **then**
    append s to *practice*
return $L = (cards, practice)$

$\bigcup_{p \in E_v} \tau(\mathrm{tgt}_p) \subseteq \bigcup_{p \in C} \tau(\mathrm{tgt}_p) = A_C$. Hence $t \in A_C$. In either case $t \in A_C$, so t occurs in some cited phrase of C and is, by Definition 1, not hallucinated. As t was arbitrary, every token of s is attested, and s presents no hallucinated token. □

Corollary 1 (Laundering impossibility)**.** *Let g be an unverified (possibly invented) candidate form, $\neg\mathrm{Ver}(g)$, and let u be a token that occurs in g but in no verified form and not in A_C. Then g contributes nothing to A, and any artifact s with $u \in \tau(s)$ has $\tau(s) \not\subseteq A$, so $\mathrm{Adm}(s) = \mathsf{false}$ and s is discarded. An invented form cannot "launder" a token into the admissible language by being emitted.*

Proof. By (4), the union forming A ranges only over v with $\mathrm{Ver}(v)$; since $\neg\mathrm{Ver}(g)$, the tokens of g are not added on its account, and by hypothesis u is in neither A_C nor any verified form, so $u \notin A$. For any s with $u \in \tau(s)$ we then have $\tau(s) \not\subseteq A$, whence $\mathrm{Adm}(s) = \mathsf{false}$ by (5). The artifact is discarded by Algorithm 2. □

The *taniwha* case. Corollary 1 is exercised by an adversarial unit test. The word *taniwha* is a genuine Māori word but does *not* occur in the toy seed corpus. Citing it to a phrase that does not contain it gives $\mathrm{Ver}(\text{"taniwha"}, \{\text{mi-001}\}) = \mathsf{false}$, so its token never enters the basis A; consequently any sentence using it, such as "*kia taniwha*," has $\mathrm{Adm} = \mathsf{false}$ and is discarded. Even a real word is refused until the corpus attests it, the filter gates on *attestation*, not on truth. The self-check prints `guardrail.check:  all assertions passed`.

Proposition 1 (Monotonicity)**.** *If $C \subseteq C'$ then $A_C \subseteq A_{C'}$; the admissible language is non-decreasing in the corpus. Moreover the count of hallucinated tokens that reach a learner is identically 0 for every corpus C, independent of $|C|$.*

Proof. If $C \subseteq C'$ then $A_C = \bigcup_{p \in C} \tau(\mathrm{tgt}_p) \subseteq \bigcup_{p \in C'} \tau(\mathrm{tgt}_p) = A_{C'}$ by monotonicity of union, and since a verified item under C remains verified under the larger evidence pool, the basis is likewise non-

Algorithm 2: Attestation filter: BuildBasis and Admit

function *BuildBasis*(C, *Vocab*)**:**
  $A \leftarrow \varnothing$
  foreach $p \in C$ **do**
    $A \leftarrow A \cup \tau(\mathrm{tgt}_p)$ `// corpus attested set` A_C`, Eq.` (2)
  foreach $v = (f_v, E_v) \in Vocab$ **do**
    $T_v \leftarrow \tau(f_v)$
    if $T_v \neq \varnothing$ ***and*** $T_v \subseteq \bigcup_{p \in E_v} \tau(\mathrm{tgt}_p)$ **then**
      `//` Ver(v)`, Eq.` (3)
      $A \leftarrow A \cup T_v$
  return A

function *Admit*(s, A)**:**
  $T \leftarrow \tau(s)$
  if $T = \varnothing$ **then return** false
  foreach $t \in T$ **do**
    if $t \notin A$ **then return** false
  return true `//` Adm(s)`, Eq.` (5)

decreasing; hence any artifact admissible under C remains admissible under C'. The hallucination count is 0 for every C by Theorem 1, which makes no assumption on $|C|$. Thus enlarging the corpus can only enlarge what may be truthfully taught, while never permitting an invented token. □

Proposition 1 is the formal content of the *flywheel*: more community data yields strictly more admissible material, at a hallucination rate that stays pinned at zero. Growth and honesty are not in tension.

4.4 Complexity and determinism

Remark 1 (Complexity). Building the basis A and testing Adm runs in $O(\sum_{p \in C} |\tau(\mathrm{tgt}_p)| + |\tau(s)|)$ time, i.e. linear in the total corpus size plus the candidate length, using a hash set for A with expected $O(1)$ membership. The procedure is decidable, deterministic, and independent of the (probabilistic) model: the same candidate against the same corpus always yields the same verdict. Soundness therefore does not rest on any statistical assumption about the generator. The model may propose freely; the cost of disposing of an unattested proposal is a single linear scan.

This separation, a soft, probabilistic *proposer* and a hard, deterministic *disposer*, is the architectural crux. The model contributes *organization* (alignments, morphological hypotheses, recombinations); the filter guarantees that the *content* is the community's.

4.5 Fail-soft architecture and the flywheel

The system is engineered so that no dependency failure can produce either a crash or an unattested artifact. Three layers fail soft: the induction layer falls back from a live LLM to a verified fixture; the persistence layer falls back from MongoDB Atlas to an in-memory store; and authentication degrades gracefully. The shared TypeScript `types.ts` contract, `zod` runtime validation, forced structured

Table 1: Per-language and aggregate metrics, independently recomputed at the live `GET /api/metrics` endpoint using the engine's own tokenizer. "Fail" is the number of practice sentences containing any unattested token (expected 0); "Halluc." is the count of hallucinated tokens that reached a learner; "Cited" is vocabulary-citation coverage. No value is self-reported.

Language	Phrases	Vocab	Patterns	Cards	Practice	Fail	Halluc.	Cited
Māori (mi)	41	34	5	12	5	0	0	34/34
Cherokee (chr)	15	14	2	10	2	0	0	14/14
Total	**56**	**48**	**7**	**22**	**7**	**0**	**0**	**48/48**

tool-use, an SM-2 spaced-repetition scheduler (`srs.ts`), Web Speech pronunciation (`tts.ts`), and Vercel Blob speaker audio complete the stack. Crucially, every fallback path still routes generated artifacts through the same attestation filter, so Theorem 1 holds on every path.

The *flywheel* is the loop's growth operator. A learner contribution appends new phrases to the corpus and invalidates the cached induction, so the next visit re-derives richer vocabulary, firmer patterns, and more practice from the enlarged C. By Proposition 1 the corpus, and thus the attested basis, grows monotonically, while the hallucination count remains zero. The language's digital record compounds with use.

5 Evaluation

5.1 Methodology

The evaluation is designed to be *independent* rather than self-reported. The live endpoint `GET /api/metrics` does not trust the engine's own output: for each inducible language it loads the seed phrases and the verified fixture induction, rebuilds the attested set from scratch using *the engine's own* τ imported from `guardrail.ts` (there is no second tokenizer), and then re-tokenizes every practice sentence and tests each token against the attested set. It separately counts vocabulary items carrying at least one citation. Because the metric recomputes the safety property with the same normalization the engine enforces, the report is a check rather than a self-grade. The property is also proven directly by a dependency-free unit check, `npx tsx src/lib/engine/guardrail.check.ts`, which prints `guardrail.check: all assertions passed`, and is reproducible against either the live deployment or a local instance.

5.2 Per-language results

Table 1 reports the per-language and aggregate metrics, recomputed live at `/api/metrics`. From 56 remembered phrases across two languages, the engine induces 48 vocabulary items, 7 grammar patterns, 22 flashcards, and 7 practice sentences. The two safety quantities are the point of the table: the number of hallucinated tokens that reached a learner is 0, and vocabulary citation coverage is 48/48. Of the 7 practice sentences checked, 0 fail attestation, i.e. 7/7 are fully attested. Figure 2 visualizes both the output volume and the safety band.

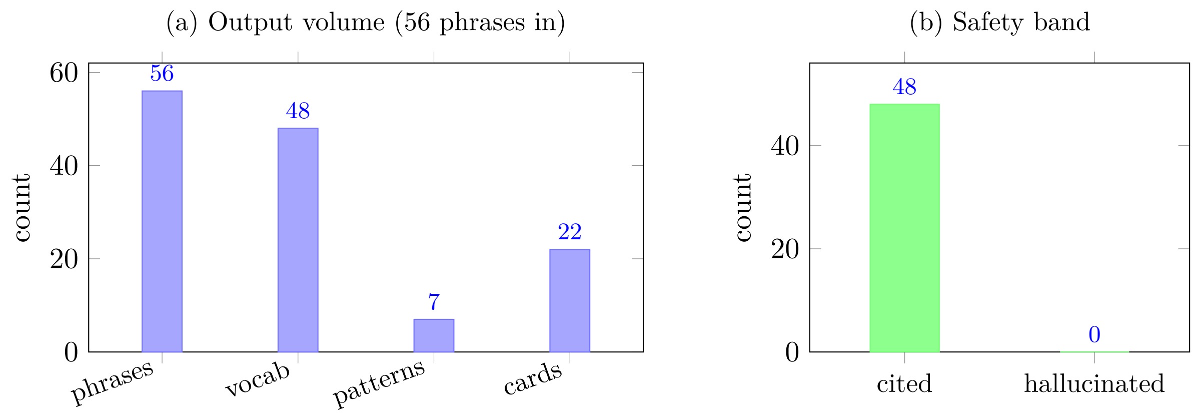

Figure 2: (a) Output volume induced from the two cited seed corpora: 56 phrases in yield 48 vocabulary items, 7 grammar patterns, and 22 flashcards. (b) The safety band: 48/48 vocabulary items carry a citation, while the count of hallucinated tokens reaching a learner is 0. Both panels are computed from the live metrics endpoint (Table 1).

5.3 Māori case study: a recovered tense system

From 41 Māori (te reo Māori) phrases the engine recovers a productive, fully cited tense-aspect system carried entirely by pre-verbal particles on the verb *haere* ("go"): *i* (past), *kei te* (present), *e* . . . *ana* (continuous), *ka* (future / inceptive), and *kua* (perfect). It further recovers that grammatical number is marked on the article (*te* singular vs. *ngā* plural) rather than on the noun; that possessive number is signaled by *t*-dropping (*tāku* "my (sg.)" vs. *āku* "my (pl.)"); and that plurality surfaces both by macron lengthening and by suppletion (*matua*/*mātua*, *tamaiti*/*tamariki*). From these attested patterns it generates new, fully attested practice such as *Ka noho au* ("I will stay"). Every token of every such sentence lies in A_C by Theorem 1; the continuous frame *e* . . . *ana* illustrates why the gap tokens of (1) are removed, the discontinuity is notation, and only the lexical material is required to be attested.

5.4 Cherokee contrast: honest low-resource degradation

From 15 isolated Cherokee (Tsalagi) items the engine recovers a vocabulary (14 items) but *honestly reports* little productive grammar (2 patterns). This is the correct low-resource behavior, not a failure: fifteen isolated words do not exhibit the minimal pairs needed to induce paradigms, and an honest system should say so rather than invent structure. The contrast between the two languages is itself the argument. Structure is non-decreasing in data (Proposition 1), Māori's 41 phrases yield 5 patterns where Cherokee's 15 items yield 2, while invention is identically 0 in both cases (Table 1). More data, more structure; never invention.

5.5 Adversarial test: the *taniwha* rejection

The adversarial test of Section 4.3 is realized as a unit check. The genuine but unattested word *taniwha* is offered with a citation that does not contain it; verification fails (Eq. 3), the token never enters the basis, and the candidate sentence is discarded by Adm. By Corollary 1 this holds for *any* token unique to an unverified form: the filter rejects 100% of unattested candidates *by construction*, not by empirical tuning. The guarantee is structural, and the self-check (`guardrail.check.ts`) certifies it on every run.

6 Discussion

Why it generalizes. LANTERN's guarantee depends on the community's cited data, not on the model's memory of any particular language. Nothing in Section 3 or in the proof of Theorem 1 is specific to Māori or Cherokee; the same construction applies to any language for which a seed corpus of glossed phrases exists. A language with a single elder and a phone can therefore begin to teach itself the same day, because the bottleneck is shifted from *model coverage* (which the long tail will never have) to *community attestation* (which a community can supply incrementally).

Originated vs. restructured knowledge. It is useful to separate the two kinds of information the system handles. In an information-theoretic, minimum-description-length framing, the model supplies the *code*, the organization that compresses the corpus into paradigms, alignments, and recombinations, while the corpus supplies the *content*, the lexical material itself. The attestation filter enforces exactly this division: it permits the model to contribute structure of arbitrary sophistication, but it forbids the model from contributing a single lexical token that the community has not attested. Hallucination, in this view, is precisely the model smuggling content into what should be only code; the filter makes that smuggling impossible (Corollary 1).

The constraint that looks limiting is the unlock. A natural objection is that forbidding the model from introducing new words is crippling. The opposite is true. The constraint is what makes the output *trustworthy enough to use* in a setting where errors are unrecoverable, and trustworthiness, not fluency, is the binding constraint on adoption for endangered-language work. By giving up the freedom to invent, the system gains a property no unconstrained generator has: a learner can rely on every word.

7 Limitations and Ethics

Pronunciation. Web Speech voices rarely exist for these languages, so synthesized playback is an approximate guide and the interface says so. Speaker-recorded audio is the right long-term answer, and the storage layer already accommodates it.

Honest low-resource grammar. With a tiny corpus the engine finds vocabulary but little grammar (Section 5.4). This is the correct behavior, and the flywheel (Proposition 1) is the remedy: structure appears as the corpus grows.

Data sovereignty. Indigenous-language data carries real ownership and governance obligations. The LANTERN stance is that communities own and govern their corpus and wield the engine on their own data; it is a tool, not an extraction pipeline. Formal sovereignty controls aligned with the CARE principles for Indigenous data governance (Carroll et al., 2020), and complementary OCAP-style ownership, control, access, and possession norms, are the first roadmap item. The evaluation here is behavioral; a full linguistic-accuracy study requires fluent-speaker review, which is the appropriate next step.

Not a replacement for speakers. LANTERN does not replace speakers, elders, or immersion. It lowers the cost of the materials *around* them, so that the scarce human time of those who still hold a language is spent teaching rather than transcribing.

8 Conclusion

Most "AI for X" systems ask a model to *be* an expert. LANTERN asks the opposite: it constrains the model into an honest amplifier and places the human community at the center of the loop. We have made that inversion precise, a decidable attestation predicate over a citation-verified corpus (Section 3), and proven that it yields zero-hallucination soundness (Theorem 1), laundering impossibility (Corollary 1), and monotone growth at a hallucination rate pinned to zero (Proposition 1), all in linear, deterministic, model- independent time (Remark 1). On two cited corpora the deployed system turns 56 remembered phrases into a complete beginner course, 48 vocabulary items, 7 patterns, 22 flashcards, 7 practice sentences, with 0 hallucinated tokens reaching a learner and 48/48 vocabulary items cited, independently recomputed.

The civilizational stake is that each language is a distinct hypothesis about what a mind can be, and losing one is losing an irreplaceable hypothesis, not a redundant copy. The bet of this work is that the cheapest and most scalable way to keep those hypotheses alive is to amplify the people who still hold them, honestly, at the speed of software. A language only dies when the last person stops speaking it; LANTERN is an attempt to make the next speaker easier to find.

Reproducibility

All reported metrics are recomputed live and are reproducible from a clean checkout (repository: github.com/areebhirani-beep/lantern; live system: lantern-cyan.vercel.app):

- `curl https://lantern-cyan.vercel.app/api/metrics`, the live deployment;
- `npm run dev` then `curl http://localhost:3000/api/metrics`, a local instance;
- `npx tsx src/lib/engine/guardrail.check.ts`, the dependency-free unit proof, which prints `guardrail.check: all assertions passed`.

The expected proof band, identical across all three, is *0 invented · 48/48 cited · 7/7 attested*.

Artifact map. Every formal construct in this paper is a faithful description of code that ships in the repository. Table 2 maps each object to its exact source location so that each claim can be verified against the implementation rather than taken on trust.

Table 2: Mapping from the paper's formal constructs to their source location in the Lantern repository (`github.com/areebhirani-beep/lantern`). Line numbers are at time of writing.

Formal object (this paper)	**Source location (repository)**
Normalization ν, tokenizer τ, gap set (Eqs. (1))	`src/lib/engine/guardrail.ts:12-27`
Corpus attested set A_C (BUILDCORPUSTOKENS)	`src/lib/engine/guardrail.ts:30-34`
Verification predicate Ver (ISVOCABVERIFIED), Eq. (3)	`src/lib/engine/guardrail.ts:41-54`
Admission predicate Adm (ISFULLYATTESTED), Eq. (5)	`src/lib/engine/guardrail.ts:57-60`
Attestation basis A and the filter (Algorithm 2, ASSEMBLE)	`src/lib/engine/index.ts:113-141`
Induction pipeline (Algorithm 1, RUNINDUCTION)	`src/lib/engine/index.ts:172-207`
Schema validation of model output (zod OUTPUTZ)	`src/lib/engine/index.ts:27-75`
Independent metric recomputation (§5.1)	`src/app/api/metrics/route.ts:13-85`
Soundness self-check / *taniwha* test (Cor. 1)	`src/lib/engine/guardrail.check.ts:49-61`
Provider-agnostic model access	`src/lib/llm.ts:14-53`
Fail-soft store (MongoDB → in-memory, §4.5)	`src/lib/store.ts:211-221`

Seed-corpus provenance. The two seed corpora are drawn from cited community and reference sources and are treated as ground truth: the Te Aka Māori Dictionary; the Digital Archive of Indigenous Language Persistence (DAILP) and the Cherokee Nation Language Department; and Omniglot phrase and number tables. These are data provenances, not scholarly references.

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